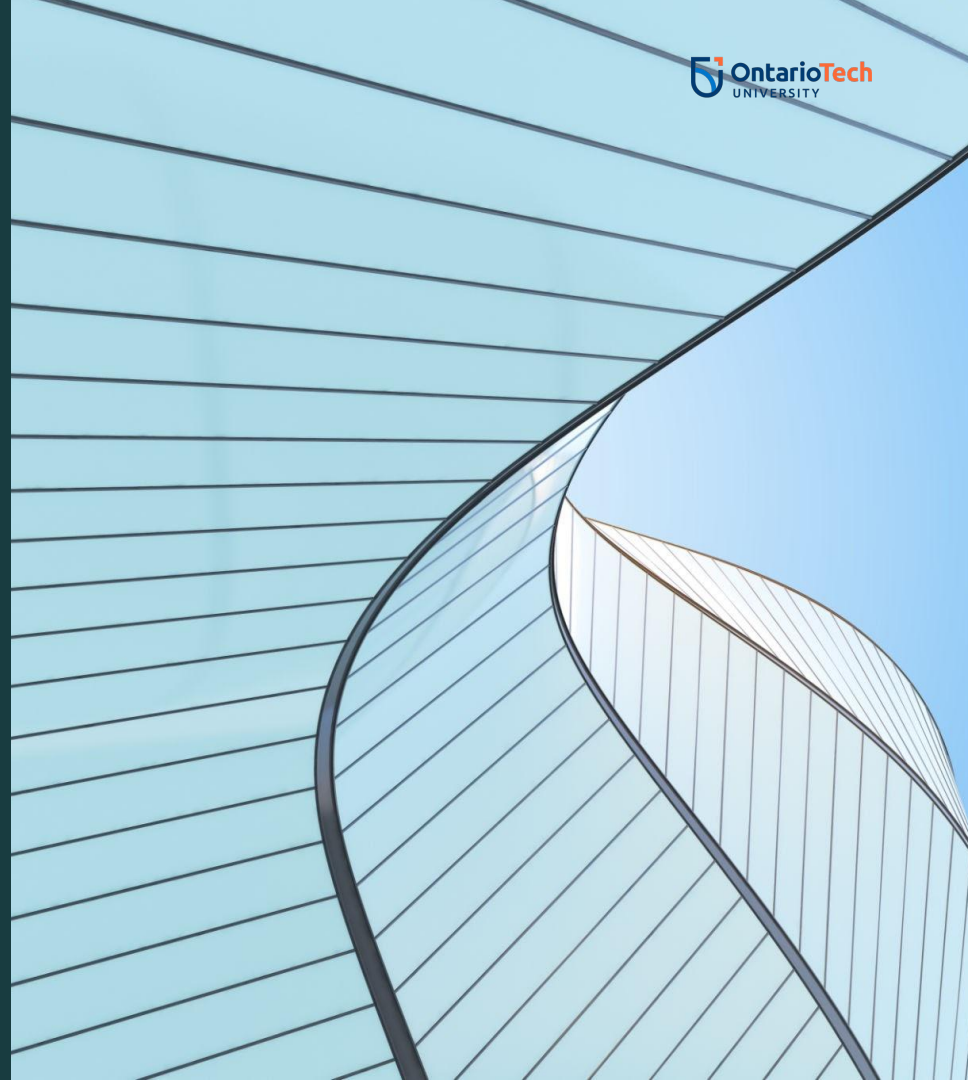


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SmartHospital Analytics Project

Case Background, Data Package and Project Objectives

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Case Background and Objectives

- SmartHospital conducted a data analytics study to evaluate patient flow and service efficiency.
- Goal: Identify factors affecting patient wait times and length of stay.
- Combined patient, laboratory, physician, and department data into a single analytics database.
- Used statistical analyses (descriptive statistics, ANOVA, correlation, regression) to uncover operational trends.
- Findings have been made to support recommendations to reduce delays, improve resource allocation, and enhance patient care.



Hospital Operations

Combined **4 SmartHospital datasets** into one analytical database:

- **Patient Visits** – demographics, wait times, length of stay, triage
- **Lab Results** – laboratory testing information
- **Physicians** – physician characteristics and experience
- **Departments** – department operations

Linked datasets using primary keys: **Visit_ID**, **Physician_ID**, and **Department_ID**.



- **Imported and merged** datasets using SAS (PROC SQL).
- **Created analytical variables** with SAS functions.
- **Performed descriptive statistics** and visualizations.
- **Conducted hypothesis tests, ANOVA, correlation, and regression.**

Graphical Analysis – Wait Times and Length of Stay

The average waiting time for patients was 42.96 minutes, which is slightly below the hospital's established standard of 45 minutes.

The average total length of stay was 188.86 minutes, meaning that patients spend over three hours at the hospital from arrival until discharge or admission.

HOSPITAL WAIT TIME SUMMARY:

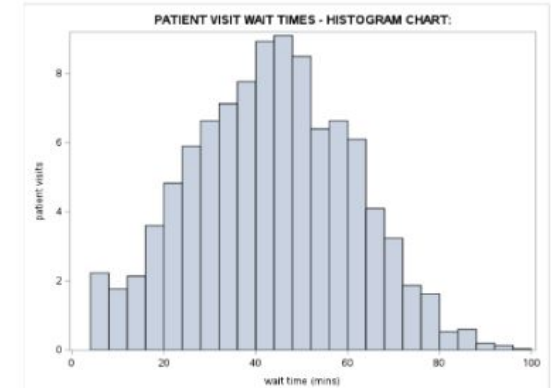
The MEANS Procedure

Variable	Label	Mean	Median	Minimum	Maximum	Std Dev
Age	Age	42.7520000	44.0000000	1.0000000	92.0000000	23.9809826
Wait_Time_Minutes	Wait_Time_Minutes	42.9580000	43.0000000	6.0000000	98.0000000	17.5462823

Program Summary - Group2_Project.sas

Variable	Label	Mean	Median	Minimum	Maximum	Std Dev
Treatment_Time_Minutes	Treatment_Time_Minutes	88.8140000	89.0000000	28.0000000	157.0000000	19.2865078
Total_Length_of_Stay_Minutes	Total_Length_of_Stay_Minutes	188.8590000	187.0000000	88.0000000	301.0000000	40.2432722
Lab_Delay_Minutes	Lab_Delay_Minutes	24.9876667	25.0000000	8.0000000	48.0000000	6.5501004
Number_of_Tests	Number_of_Tests	3.0030000	3.0000000	1.0000000	5.0000000	0.8305947

Figure 1: Distribution of Patient Wait Times



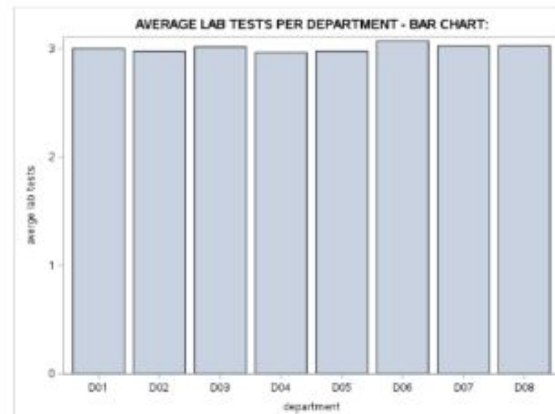
Department Patient Volume

Department Workload & Laboratory Demand Summary

- Patient visits are unevenly distributed across departments.
- Emergency Medicine (D01) received the highest number of visits (26.0%), while Gastroenterology (D08) handled the lowest (6.47%).
- Unequal patient distribution may increase workload pressure and lead to longer wait times.
- Laboratory test volumes vary by department, reflecting differences in patient needs, although all departments require very similar demand for Lab tests on a regular basis.
- High-demand departments may require additional staff and laboratory resources to maintain efficient patient care.

Department_ID		
Department_ID	Frequency	Percent
D01	780	26.00
D02	328	10.93
D03	334	11.13
D04	280	9.33
D05	515	17.17
D06	315	10.50
D07	254	8.47
D08	194	6.47

Average Number of Lab Tests per Department





Graphical Analysis – Triage Levels and Patient Disposition

Triage Distribution: Level 3 was the most common (34.93%), followed by Level 4 (28%), while Level 1 (8.03%) represented the fewest high-priority cases.

Patient Outcomes: 49.20% of patients were admitted, 43.83% were discharged, and 6.97% were transferred.

The large number of Level 3 and 4 patients, combined with the high admission rate, highlights the need for adequate staffing, inpatient bed capacity, and efficient resource allocation.

Disposition				
Disposition	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Admitted	1476	49.20	1476	49.20
Discharged	1315	43.83	2791	93.03
Transferred	209	6.97	3000	100.00

Triage_Level				
Triage_Level	Frequency	Percent	Cumulative Frequency	Cumulative Percent
1	241	8.03	241	8.03
2	536	17.87	777	25.90
3	1048	34.93	1825	60.83
4	840	28.00	2665	88.83
5	335	11.17	3000	100.00

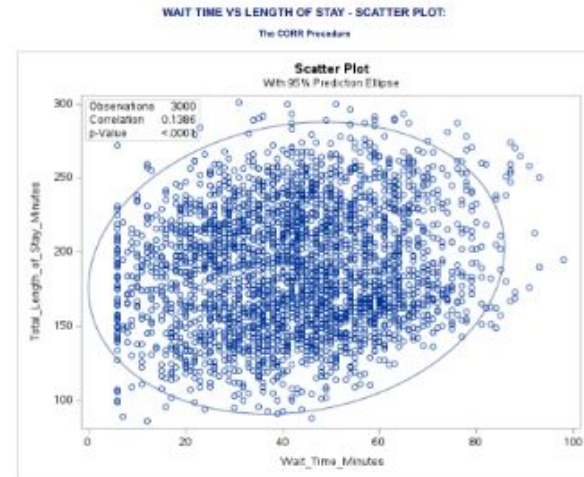
Graphical Analysis – Average Shift Wait Time and Wait Time vs. Length of Stay**AVERAGE WAIT TIME PER SHIFTS SUMMARY:**

	Wait_Time_Minutes
	Mean
Shift	
Afternoon	45.49
Morning	35.04
Night	52.17

Morning had the shortest average wait (35 min); night had the longest (52 min).

Wait time and length of stay showed a **weak positive correlation** ($r = 0.1386$).

Staffing and patient volume likely have a greater impact on hospital efficiency than wait time alone.

Wait Time vs. Length of Stay (Figure 3)

Hypothesis Tests

- **One-Sample t-Test:** Average wait time (**42.96 min**) was significantly **lower** than the 45-minute benchmark ($p < 0.001$), indicating the hospital met its target.
- **Two-Sample t-Test:** Weekend wait times (**46.45 min**) were significantly longer than weekday wait times (**41.44 min**) ($p < 0.001$).
- The hospital performs better than its wait-time benchmark overall, but weekend demand leads to longer patient waits.

AVERAGE WAIT TIME VS 45-MINUTE BENCHMARK- ONE SAMPLE TTEST:

The TTEST Procedure

Variable: Wait_Time_Minutes (Wait_Time_Minutes)

N	Mean	Std Dev	Std Err	Minimum	Maximum
3000	42.9560	17.5463	0.3203	6.0000	98.0000

Mean	95% CL Mean	Std Dev	95% CL Std Dev
42.9560	42.3279 43.5841	17.5463	17.1133 18.0019

DF	t Value	Pr > t
2999	-6.38	<.0001

COMPARING WAIT TIME FOR WEEKDAY AND WEEKEND VISITS - INDEPENDENT TWO SAMPLE TTEST:

The TTEST Procedure

Variable: Wait_Time_Minutes (Wait_Time_Minutes)

Weekend_Flag	Method	N	Mean	Std Dev	Std Err	Minimum	Maximum
0		2094	41.4422	17.4192	0.3807	6.0000	93.0000
1		906	46.4547	17.3473	0.5763	6.0000	98.0000
Diff (1-2)	Pooled		-5.0125	17.3978	0.6918		
Diff (1-2)	Satterthwaite		-5.0125		0.6907		

Weekend_Flag	Method	Mean	95% CL Mean	Std Dev	95% CL Std Dev
0		41.4422	40.6957 42.1887	17.4192	16.9072 17.9635
1		46.4547	45.3237 47.5858	17.3473	16.5837 18.1852
Diff (1-2)	Pooled	-5.0125	-6.3690 -3.6500	17.3978	16.9862 17.8494
Diff (1-2)	Satterthwaite	-5.0125	-6.3672 -3.6578		

Method	Variances	DF	t Value	Pr > t
Pooled	Equal	2998	-7.25	<.0001
Satterthwaite	Unequal	1724.9	-7.26	<.0001

Equality of Variances				
Method	Num DF	Den DF	F Value	Pr > F

One Way Anova Results

One-Way ANOVA: Patient Wait Time by Departments (Minutes)

The ANOVA Procedure

Dependent Variable: Wait_Time_Minutes Wait_Time_Minutes

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	7	89120.5024	12731.5003	45.66	<.0001
Error	2992	834187.6896	278.8060		
Corrected Total	2999	923308.1920			

R-Square	Coeff Var	Root MSE	Wait_Time_Minutes Mean
0.096523	38.87114	16.69749	42.95600

Source	DF	Anova SS	Mean Square	F Value	Pr > F
Department_Name	7	89120.50237	12731.50034	45.66	<.0001

One-Way ANOVA: Wait times differed significantly across the eight hospital departments ($p < 0.001$).

Department Effect: Department accounted for about **9.65%** of the variation in patient wait times.

Post Hoc Results: All three shifts (Morning, Afternoon, and Night) had significantly different wait times ($p < 0.001$).

Key Finding: Morning had the shortest wait time (34.75 min) and Night the longest (50.18 min), suggesting additional staffing may be needed during the Night shift.

Two-Way Anova

Two-way Anova was conducted to evaluate the effectiveness of Department ID and Shifts on patient wait time.

Department ID

Significant effect on patient wait time ($p < 0.0001$); average wait times differed across departments.

Shift

Significant effect on patient wait time ($p < 0.0001$); wait times varied depending on the shift.

Department ID and Shift

Significant interaction ($p < 0.0001$); the effect of shift on wait time differed by department, with some departments experiencing much larger changes than others.

Model	23	234999.7751	10217.3815	44.18	<.0001
Error	2976	688308.4169	231.2864		
Corrected Total	2999	923308.1920			

R-Square	Coeff Var	Root MSE	Wait_Time_Minutes Mean
0.254519	35.40391	15.20810	42.95600

Source	DF	Type I SS	Mean Square	F Value	Pr > F
Department_ID	7	89120.5024	12731.5003	55.05	<.0001
Shift	2	134111.6250	67055.8125	289.93	<.0001
Department_ID*Shift	14	11767.6477	840.5463	3.63	<.0001

Source	DF	Type III SS	Mean Square	F Value	Pr > F
Department_ID	7	83341.12053	11905.87436	51.48	<.0001
Shift	2	94829.52802	47414.76401	205.00	<.0001
Department_ID*Shift	14	11767.64765	840.54626	3.63	<.0001

Recommendations

1. Increase Night Shift Staffing

Allocate additional staff and resources during night shifts to reduce longer patient wait times.

2. Increase High-Demand Departments

Optimize workflows and resources in departments with the highest patient volumes and wait times.

3. Reduce Treatment and Lab Delays

Streamline treatment processes and laboratory services to shorten patients' overall length of stay.

4. Monitor Performance with Data Analytics

Continuously track key metrics, including wait times, length of stay, and departmental workload, to identify and address issues early.

5. Enhance Patient Flow

Standardize and improve operational processes to reduce delays, increase efficiency, and improve the overall patient experience.

Limitations

- Single Hospital Dataset: Findings may not be generalizable to other hospitals or healthcare settings.
- Limited Variables: Important factors such as staffing levels, bed availability, patient volume, and seasonal trends were not included.
- Weak Relationships: Some variables showed weak associations, suggesting other unmeasured factors may influence hospital performance.
- Observational Study Design: The use of historical and observational data identifies associations but cannot establish cause-and-effect relationships.

Conclusion

SmartHospital met overall wait time targets, but differences across departments and shifts reveal opportunities to improve patient flow and operational efficiency.

Key learning 1

Patient wait times vary significantly by department and shift, with night shifts and weekends experiencing the longest delays.

Key learning 2

Treatment time, laboratory delays, triage duration, and patient priority are the primary factors influencing patients' length of stay.

Key learning 3

Data analytics is an effective tool for identifying operational inefficiencies and supporting continuous improvements in patient flow and healthcare service delivery.

THANK
YOU